

Unsupervised Discovery of Persona-Relevant Directions with Matryoshka Sparse Autoencoders

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Abstract. Persona vectors—activation directions that control traits such as sycophancy or dishonesty—are found by *naming* a trait, writing contrastive prompts, and differencing activations, which requires the trait to be specified in advance, precisely described, and prompt-inducible. Chen et al. (2025) conjecture that “SAEs may therefore enable unsupervised discovery of persona-relevant directions, including specific traits that cannot be easily elicited through prompting.” We take up that conjecture at scale: we train a Matryoshka BatchTopK sparse autoencoder (235M parameters, 32,768 features) on 100M assistant-position tokens of Qwen2.5-7B-Instruct at layer 19 (held-out FVE 0.72, $L_0=40$), rank its dictionary for persona-relevance with three label-free arms, steer the model along each candidate, and screen the steered behaviour against random-direction nulls and persona-vector *controls* before a blinded judge names the survivors—no trait specified and no description written at any point. In a powered screen (24 scenarios), the method validated against **2 of 3** supervised persona-vector controls, passed **60 of 128** SAE features through the full gate, and a blinded judge assigned **10** of them consistent, above-null names—recognisable persona axes such as *engaged and supportive*, *empathetic*, *formal/academic*, *playful*, and *composed and confident*—with no trait ever specified. (The evil control stayed just under the consistency gate, so the count is conservative.) We do *not* claim these directions are unreachable by prompting; measuring the *graded* form of Chen et al.’s “easily”—how hard a prompt must work to reproduce each—is future work.

At a glance

- **Dictionary:** 235M-param Matryoshka BatchTopK SAE, width 32,768, $k=40$, nested 1k/4k/16k/32k shells.
- **Trained on:** 100M assistant-position tokens of Qwen2.5-7B-Instruct, layer 19 → **held-out FVE** 0.72, $L_0 = 40$.
- **Discovery:** three *label-free* ranking arms; no trait ever specified.
- **Screen:** steering + separability AUC vs 20 random nulls, gated by persona-vector controls; blinded LLM judge names survivors.
- **Drops requirements 1–2** of prompt-based extraction: no trait specified, no description written.
- **Open question (future work):** *how easily* a prompt can reproduce each discovered direction—a graded elicitation cost, not a “prompt-unreachable” claim.
- **Result:** 2/3 controls validate the screen; 60/128 features clear the gate; **10 persona directions discovered unsupervised and named** (warmth, empathy, formality, playfulness, confidence).

Infrastructure. Run end-to-end on a reproducible harness (durable workflows, frozen protocol hashes, a spend ledger and cost caps); the SAE is trained once on an A100 and reused off a Hugging Face checkpoint for every screen. **Builds on** R. Chen, A. Arditì, H. Sleight et al., *Persona Vectors: Monitoring and Controlling Character Traits in Language Models*, arXiv:2507.21509 (2025), whose appendix first suggests this SAE direction.